

# Threat Models and Repetition Windows for Anonymous DHTs: A Short Declaration Checklist for Referee-Grade Budget Claims

Unified archive note v0.22 (2026-02-28; archive v149)

## Abstract

Anonymous DHT papers are easy to read past each other because “the threat model” hides several independent choices: what the secret is (initiator vs target), what transcript projection is visible (tier), what the repetition unit/window is, and how much observation/manipulation power the adversary has. This short note standardizes a small declaration checklist meant to be cited (not re-derived) by the Boss-Fight dial sheet, certified-menu selection, and evaluation/notarization papers. It is complementary to the series trace-interface note, which owns the transcript-projection vocabulary.

**A short “TW declaration” template.** At minimum, a referee should be able to locate these fields: secret (*initiator* vs *destination key*), observation tier (by `exposure_nf_id`), repetition unit, window, adversary control (passive vs selective-failure vs full Byzantine), and background conditioning (public calendars, committee partitions). This note treats that tuple as the named object `TW` whose digest is `tw_id`.

**Remark 1** (A concrete template instance). *Synthesis 17’s worked example instantiates this template with the label `tw.reader_24h.q500.v1` and a digest-bound `tw_id` that is reused across multiple receipt line items. The point is mechanical: a referee should be able to diff the window declaration (and its conditioning universe) without reading prose.*[5, 4]

**Scope.** Standardize threat-model and repetition-window declarations for Anonymous-DHT anonymity budgets (so papers can stay non-redundant).

**Imports.** We import transcript-projection vocabulary (observation tiers / exposure normal forms) from the trace-interface synthesis note.[2] Receipt line items bind these declarations via digest IDs (`exposure_nf_id`, `tw_id`); see Synthesis 12.[4] Primary-path role-separation assumptions (and their optional probabilistic adjuncts) are imported from the separation-manifest notes.[6, 7]

**Exports.** A concise checklist (and named tuple) for: secret scope, visible tier, repetition unit/window, conditioning universe, and adversary powers (passive/active/background).

**Artifact policy.** Source-only; supporting artifacts are available on request. See [1].

# 1 What this note owns (non-redundancy)

The series trace-interface note owns *what is visible* (observation tiers, exposure normal forms, and the CCT schema) [2]. This note owns *how long* and *under what repetition*: (i) the repetition unit/window declaration; (ii) the candidate universe and conditioning rules that define the secret scope; and (iii) a minimal adversary-power menu (passive, active/selective-failure, background-manipulation). Other papers should cite this note instead of repeating generic “Boss Fight vs passive” paragraphs.

## 2 A minimal declaration checklist

Any referee-facing anonymity claim in this archive should explicitly declare the following fields. (These fields are also natural inputs to ABOM/OINL certificates.)

**Definition 1** (Threat/window declaration). *A threat/window declaration is a tuple*

$$TW = (\text{secret}, \text{tier}, \text{unit}, \text{window}, \text{universe}, \text{obs}, \text{active}, \text{bg}).$$

*The fields are:*

1. **Secret.** *What is being protected: initiator identity  $I$ , lookup target/key  $K$ , destination committee label  $A \in \mathcal{A}$ , or a profile/class  $C$ .*
2. **Tier.** *Which transcript projection is being bounded (Tier 0–3) and its exposure normal form [2].*
3. **Unit.** *The repetition unit for composition (per lookup, per epoch, per round-trip, per retry attempt).*
4. **Window.** *The window length/horizon for the claimed budget (e.g.,  $T$  epochs, or “per 24h sliding window”) and the reset/checkpoint rule.*
5. **Universe.** *The candidate universe size and conditioning: e.g.,  $|\mathcal{A}|$  committees at Tier 0, or an effective target set size  $K$  after public side information, caching, and eligibility disclosures.*
6. **Observation model.** *A declared observation/compromise parameterization appropriate to the tier (e.g.,  $p_{\text{obs}}$  per contact at Tier 2; or “observer sees a single label per epoch” at Tier 0).*
7. **Active power.** *What deviations are in-scope: selective failures, induced retries/delays, Sybil/route capture, equivocation in logged receipts.*
8. **Background assumptions.** *If the claim references a public background  $Q$  (equalization), declare how  $Q$  is produced (specified vs measured), how drift is detected, and what manipulation is in-scope.*

**Deployed control-plane.** If *obs* or *bg* are defined in terms of delegated-routing endpoints, AutoConf, gateway defaults, or other IPFS/libp2p control-plane knobs, treat those knobs as part of the declared state and see Synthesis 30 for a compact “what-to-pin” checklist.[3]

**Definition 2** (Threat-window id (TW-ID)). *Like interfaces, threat/window declarations should be treated as claim fields. Define a stable identifier*

$$\text{tw\_id} = \text{SHA256}(\text{canon}(TW)) \in \{0, 1\}^{256},$$

*where  $\text{canon}(\cdot)$  is a canonical serialization of (secret, tier, unit, window, universe, obs, active, bg), including the exposure-normal-form id (ENF-ID) for tier when applicable. Publish  $\text{tw\_id}$  as `sha256:<lowercase-hex>` (explicit algorithm tag). A short prefix may be used in text provided the full digest is in the ABOM.*

**Remark 2** (Field naming (avoid case drift)). *In this archive, the claim field is written as `tw_id` (lowercase) in receipts, ABOMs, and OINLs. Mixing `TW_id` and `tw_id` as distinct keys is an avoidable source of audit drift; treat the `id` as a single field.*

**Remark 3** (Why TW-ID matters). *Without TW-ID, a certificate can silently change its composition horizon (“per epoch” vs “per day”), its candidate universe, or its permitted active deviations. TW-ID makes these choices explicit and supports comparison across releases and across papers.*

**Remark 4** (Why these fields are separable). *A Tier 0 “single-label” guarantee can be meaningful even if Tier 2 is hopeless, but only if the deployment enforces interface discipline (logs/certificates do not accidentally publish higher-tier traces). Similarly, a strong per-lookup bound is meaningless if the deployment allows adversarially induced retries that expand the repetition window. Splitting the declaration into tier and window avoids silent mismatches.*

**Remark 5** (Separation assumptions belong in the declaration). *If a claim relies on a “primary path” whose security is conditional on non-collusion (e.g., `client` → `relay` → `gateway`), the separation assumption is part of the threat model, not an implementation footnote. In this series, treat it as a digest-bound guard object (a separation-manifest `id`) and, when publishing a quantitative failure bound, as a guarded checkpoint via `rho_upper` (Synthesis 23 gives the receipt-side  $(0, \rho)$  interpretation). [6, 7]*

### 3 Two threat-model templates (used across the archive)

We intentionally keep templates few. Papers should specify deviations from these templates rather than introducing new ones casually.

**Template P (passive profiling window).** Secret is a class/identity variable  $C$ . Tier is 0 or 1. Unit is “epoch” (or lookup). Window is long ( $T \gg 1$ ). The adversary is passive and observes the declared tier projection each unit. Background  $Q$  may drift and must be logged/budgeted if it is used in a contract. This template is the default for the profiling/equalization paper (Anonymity B).

**Template B (Boss Fight window).** Secret is a target/destination  $K$  (or committee label) and the adversary can actively shape the transcript: selective failures, induced retries, and timing perturbations. Tier is typically 2 (ordered failures) and the repetition unit is “attempt” or “lookup”. The window is defined by an operational query budget  $Q$  (lookups per time window) together with a reset rule. This template is the default for the Boss-Fight dial sheet and for proof-carrying repetition accounting.

### 4 A small observation-probability lemma (scale anchor)

Many papers use a single parameter  $p_{\text{obs}}$  as a scale anchor: the probability that a relevant transcript component is observed by the adversary. The exact meaning is tier-relative and must be declared, but the following proxy is frequently adequate as a conservative anchor.

**Remark 6** (Committee contact observation proxy). *If each contacted committee is of size  $s$  sampled from  $N$  nodes and the adversary corrupts a fraction  $\rho$ , then a pessimistic sufficient condition for “observing” that committee is that it contains at least one corrupted node:*

$$p_{\text{obs}} = 1 - \frac{\binom{(1-\rho)N}{s}}{\binom{N}{s}} \approx 1 - (1 - \rho)^s.$$

*This proxy is used only as a scale anchor; real overlays exhibit correlations (eclipse/route capture).*

## How to use this note

When a paper needs “threat model” boilerplate, it should: (i) cite this note for the TW checklist; (ii) cite the trace-interface note for the tier/exposure normal form; and (iii) state only the field instantiations (secret, tier, unit, window, universe, obs, active, bg) needed for the local theorem.

## References

- [1] Anonymous. Artifact and Disclosure Policy for the One-True-Archive (Source-Only Public Release). Series synthesis note (Synthesis 6), 2026. (This archive.)
- [2] Anonymous. Trace Interfaces for Anonymous DHTs: Observation Tiers, Committee-Contact Traces, and Exposure Normal Forms. Series synthesis note, 2026. (This archive.)
- [3] Anonymous. Deployed Observation Surfaces for Anonymous DHTs: Control-plane knobs in IPFS/libp2p delegated routing and OHTTP evolution. Series synthesis note (Synthesis 30), 2026. (This archive.)
- [4] Anonymous. Receipt Line Items for Anonymity Claims: A Minimal Tuple Schema for Published Budgets. Series synthesis note (Synthesis 12), The-One-True-Archive (2026).
- [5] Anonymous. Worked Example: Instantiating a Tiered Reader-Privacy Receipt. Series synthesis note (Synthesis 17), The-One-True-Archive (2026).
- [6] Anonymous. Primary-Path Separation Manifests for Anonymous-DHT Receipts: Guard Objects for Non-Collusion Assumptions. Series synthesis note (Synthesis 22), 2026. (This archive.)
- [7] Anonymous. Probabilistic Separation Budgets: Separation Failure as a Receipt-Grade  $(\epsilon, \delta)$  Adjunct. Series synthesis note (Synthesis 23), 2026. (This archive.)
- [8] J. R. Douceur. The sybil attack. In *IPTPS*, 2002.
- [9] A. Singh, T. Ngan, and P. Druschel. Eclipse attacks on overlay networks: threats and defenses. In *IEEE INFOCOM*, 2006. (Initial versions circulated 2004.)
- [10] Q. Wang, P. Mittal, and N. Borisov. In search of an anonymous and secure lookup: attacks on structured peer-to-peer anonymous communication systems. In *ACM CCS*, 2010.
- [11] P. Mittal and N. Borisov. Information leaks in structured peer-to-peer anonymous communication systems. *ACM TISSEC*, 15(1), 2012.